András Jankó

Budapest | +36 70 415 5054 | 2000.andras.janko@gmail.com

GitHub: github.com/Pillangocska

SKILLS

- **Programming languages:** Python, Java, JavaScript, SQL
- **Frameworks and technologies:** Scikit-learn, Keras, PyTorch, Transformers, Git, SpringBoot, Docker, K8S, Grafana, RabbitMQ, SAS Viya, Apache Airflow
- Strictly following OOP principles
- Fundamental understanding of machine learning algorithms
- Languages: Hungarian (native), English (C1 certification)

EXPERIENCE

Network Operations Center Engineer intern | CETIN Hungary | 1 year (2022 – 2023)

- Monitored network infrastructure and services, responded to network alerts and incidents troubleshooted connectivity issues and service disruptions, while experiencing how a multinational environment with an agile mindset works
- Learned network protocols, routing, and switching fundamentals, worked with ticketing systems for incident management and resolution tracking, wrote and maintained scripts for operational tasks

Machine Learning Engineer | Quadron Cyber Security & Analytics Zrt. | 2.5 years (2023 – present)

- **Developed a monitoring application** for pellet heaters using SpringBoot Java backend, implemented containerization and DevOps practices, created analytics dashboards with Grafana, and built forecasting capabilities using traditional statistical methods
- **Built an anomaly detection system** analyzing vibration and structural characteristics of bridges, developing forecasting models to predict when infrastructure will degrade beyond repair
- **Participated in an international project developing a data platform** similar to SAS Viya or Databricks, using Spark, HDFS, Dask and other open source technologies to process and analyze large volumes of textual data. Contributed to both the machine learning pipeline development and platform architecture, focusing on scalable text processing and analysis capabilities for open source data sources

- **Currently aggregating and analyzing multiple government department data sources using SAS Viya** while implementing advanced NLP solutions with LLMs and statistical methods for data insights. Responsible for the complete data engineering infrastructure including designing and managing Apache Airflow pipelines, handling deployment and orchestration on Kubernetes, and ensuring scalable data processing workflows across the entire system.

University Projects

- **Master's thesis (ongoing)**

  Developing a proof-of-concept data platform using microservices architecture to process massive textual datasets and generate network structures in graph databases. Focusing on topic modeling techniques and analyzing information flow patterns within the generated networks

- **Bachelor's Thesis** - *Fine-tuning and user interface integration of large language models*

  Reference:
  https://github.com/Pillangocska/Data_Science_University_Archive
  https://diplomaterv.vik.bme.hu/hu/Theses/Nagy-nyelvi-modellek-finomhangolasa-es

- **Software Project Laboratory**

  A relatively complex 2d game in Java, that we developed in a group of 5 developers. I learned and applied design patterns, code documentation and architecture planning.
  Reference: https://github.com/Pillangocska/Alfa_Csapat_Grafikus

- **Software Architectures**

  A distributed system for real-time monitoring and control of ski resort lifts. We developed it in a group of 4 developers, used many architectural patterns and technologies.
  Reference: https://github.com/Pillangocska/SnowPeak

EDUCATION

Budapest University of Technology and Economics (BME)

Computer Engineering BSc (*2024)*

- Department of Telecommunications and Artificial Intelligence (TMIT)

Computer Engineering MSc (*2024 – present)*

- Data Science and Artificial Intelligence specialization (TMIT & MIT)